Echo Conference 2/13/25

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Inova™

Quantifying Aortic Insufficiency

- Continuity Equation
- Flow Convergence (PISA)
- 3D Echocardiography

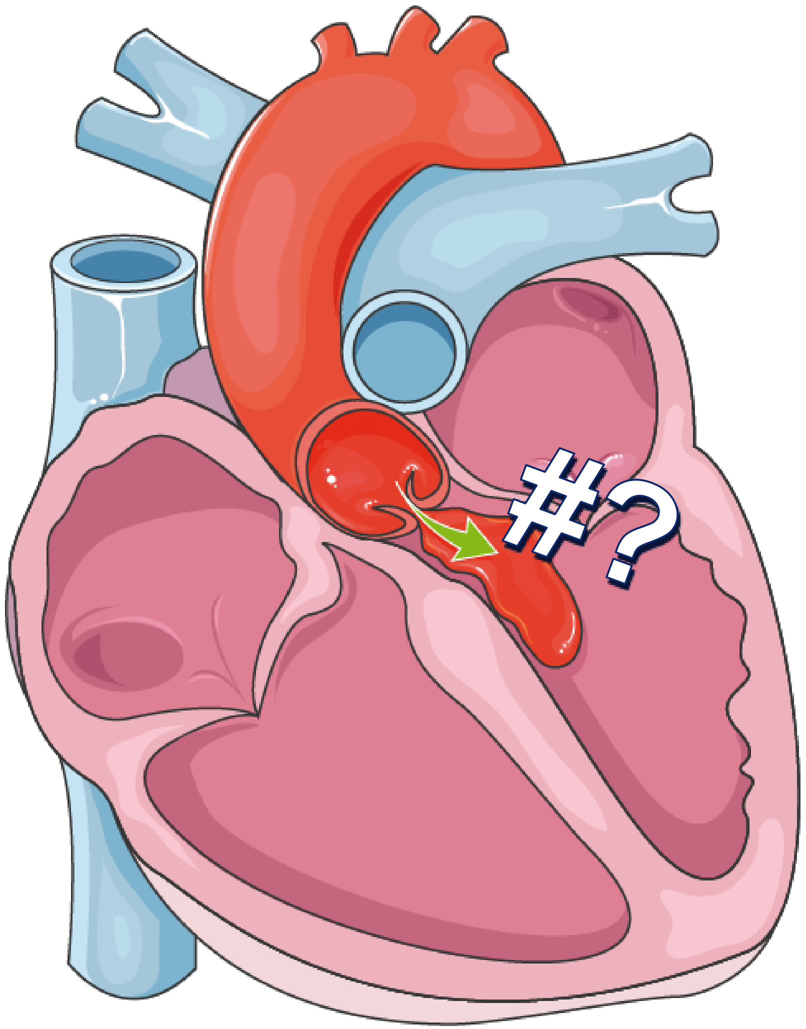
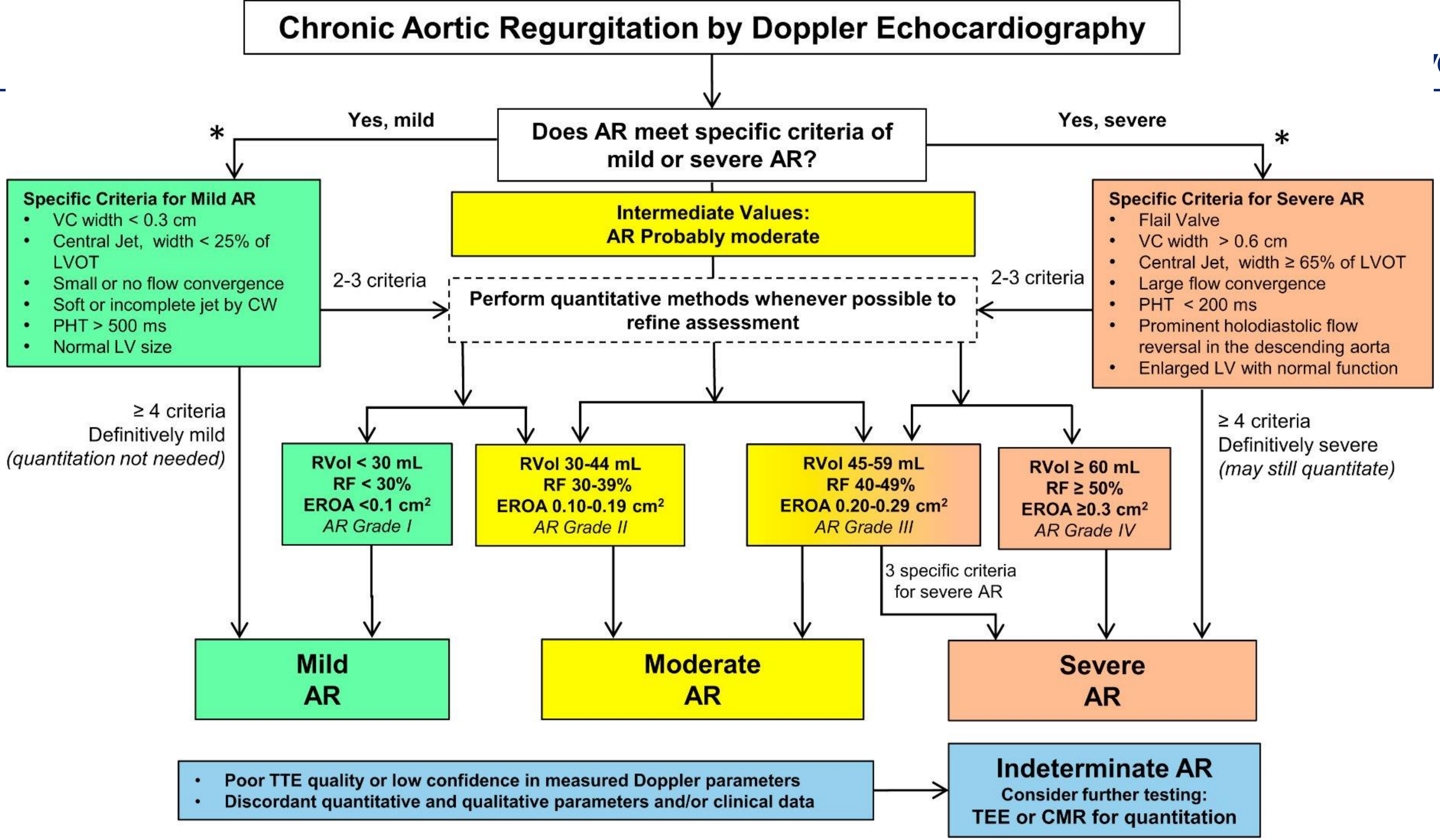
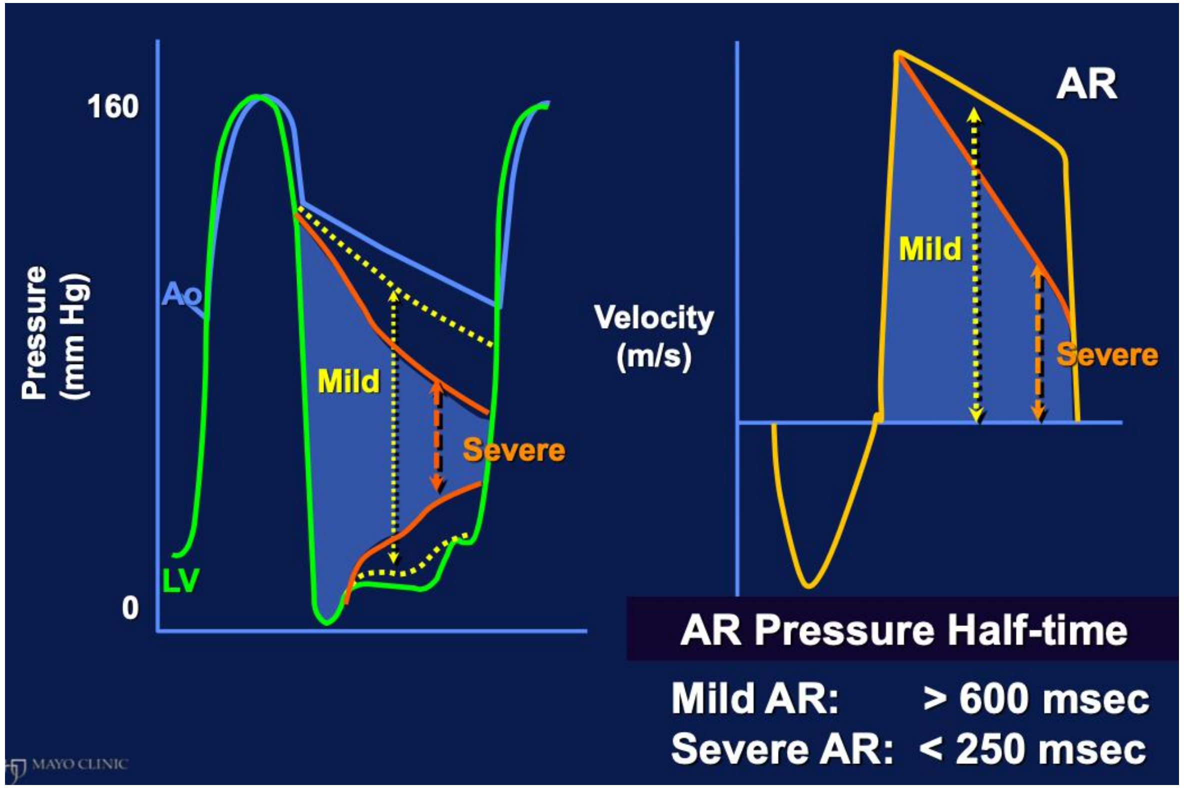


Table 11 Grading the severity of chronic AR with echocardiography

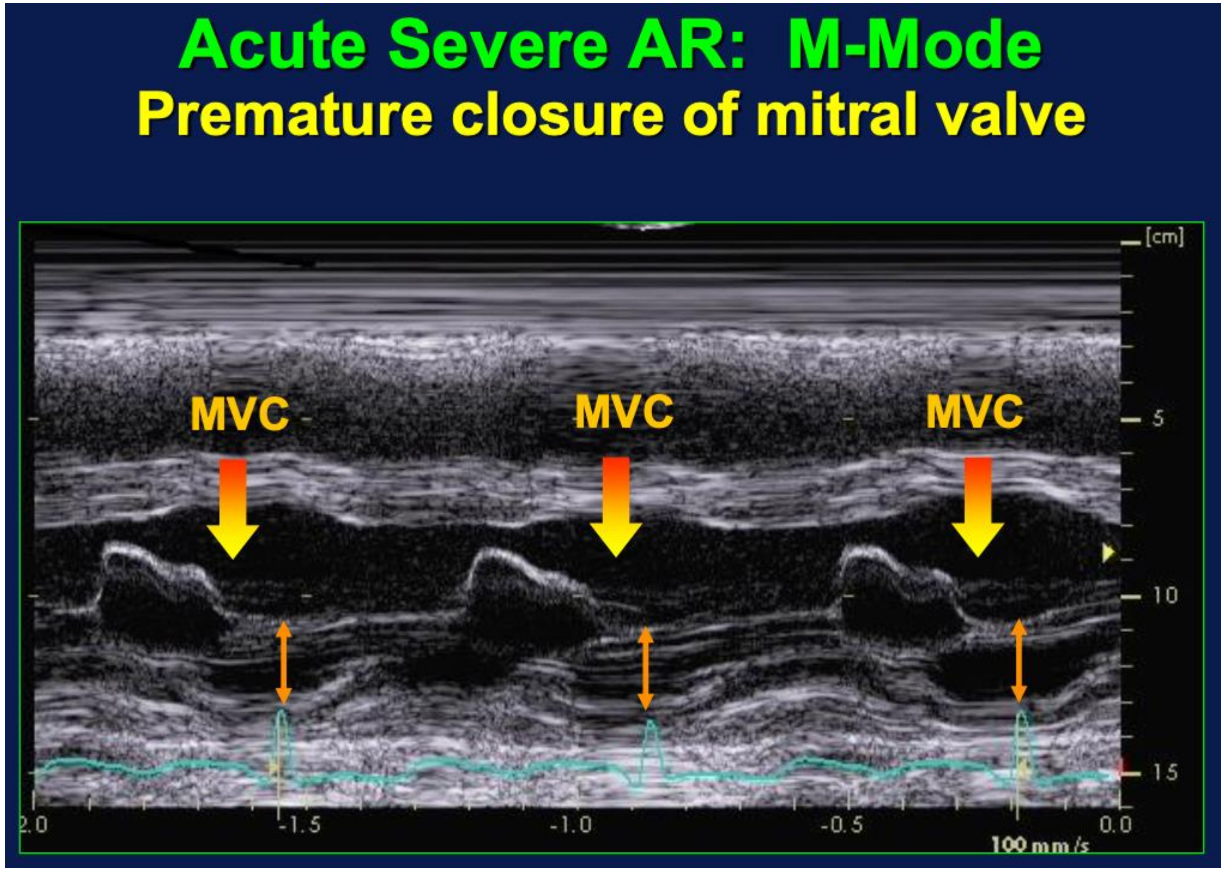
	AR severity			
	Mild	Moderate		Severe
Structural parameters				
Aortic leaflets	Normal or abnormal	Normal or abnormal		**Abnormal/flail, or wide coaptation defect**
LV size	**Normal**[*]	Normal or dilated		Usually dilated[†]
Qualitative Doppler				
Jet width in LVOT, color flow	**Small in central jets**	Intermediate		**Large in central jets**; variable in eccentric jets
Flow convergence, color flow	**None or very small**	Intermediate		**Large**
Jet density, CW	**Incomplete or faint**	Dense		Dense
Jet deceleration rate, CW (PHT, msec)[‡]	Incomplete or faint Slow, >500	Medium, 500-200		**Steep, <200**
Diastolic flow reversal in descending aorta, PW	**Brief, early diastolic reversal**	Intermediate		**Prominent holodiastolic reversal**
Semiquantitative parameters[§]				
VCW (cm)	<0.3	0.3-0.6		>0.6
Jet width/LVOT width, central jets (%)	<25	25-45	46-64	≥65
Jet CSA/LVOT CSA, central jets (%)	<5	5-20	21-59	≥60
Quantitative parameters[§]				
RVol (mL/beat)	<30	30-44	45-59	≥60
RF (%)	<30	30-39	40-49	≥50
EROA (cm^2)	<0.10	0.10-0.19	0.20-0.29	≥0.30



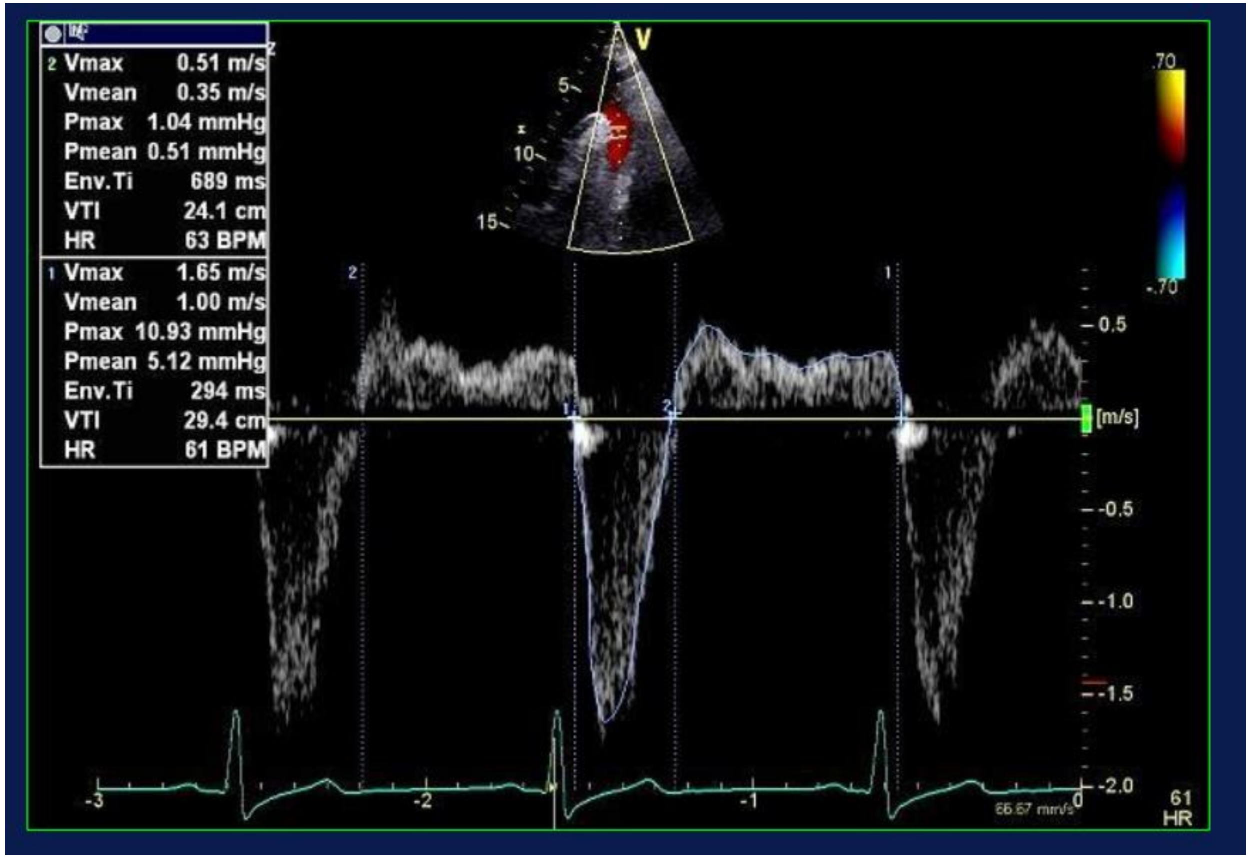
*	Beware of limitations of color flow assessment in eccentric AR jets; volumetric quantitation and integration of other parameters is advised



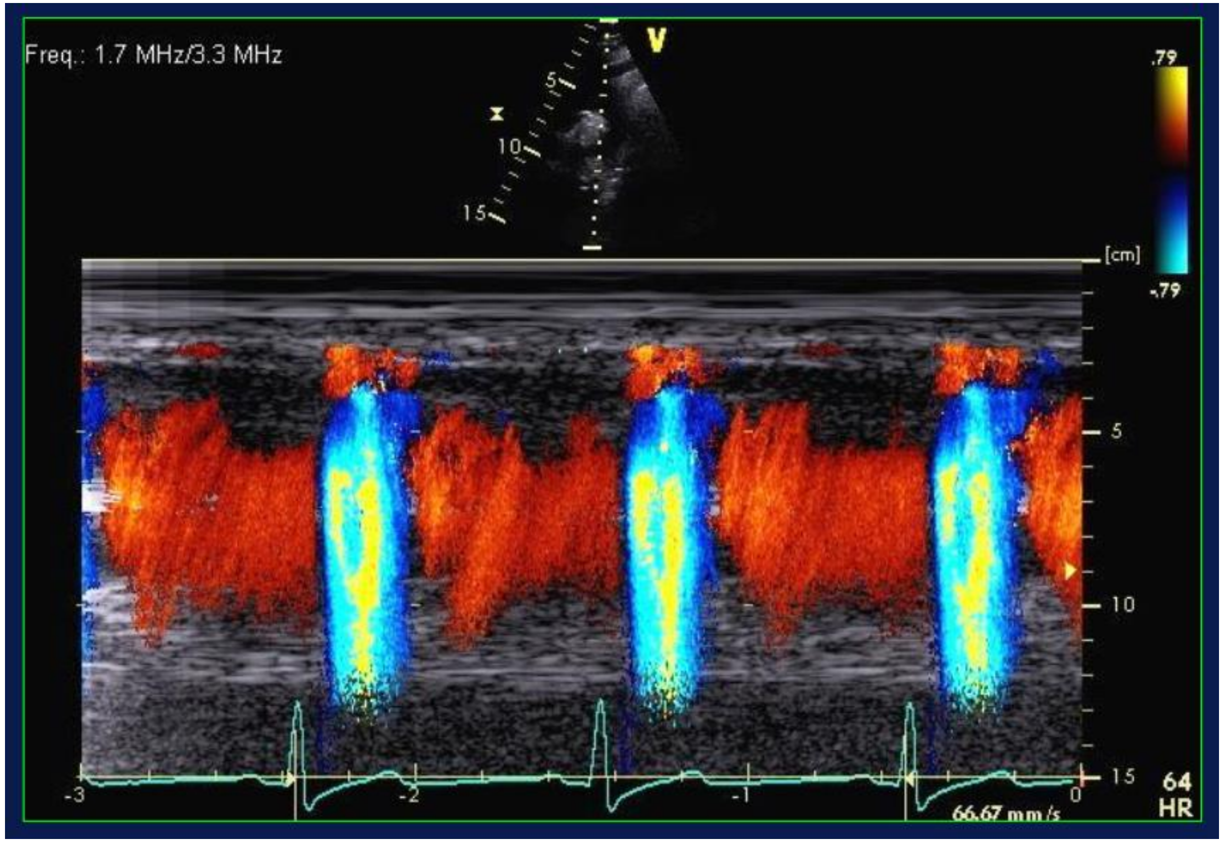
160
0
Pressure (mm Hg)
Ao
LV
Mild
Severe
Velocity (m/s)
AR
Mild
Severe
AR Pressure Half-time
Mild AR: > 600 msec
Severe AR: < 250 msec
MAYO CLINIC



Acute Severe AR: M-Mode
Premature closure of mitral valve
MVC
MVC
MVC
[cm]
5
10
15
2.0
-1.5
-1.0
-0.5
0.0
100 mm/s

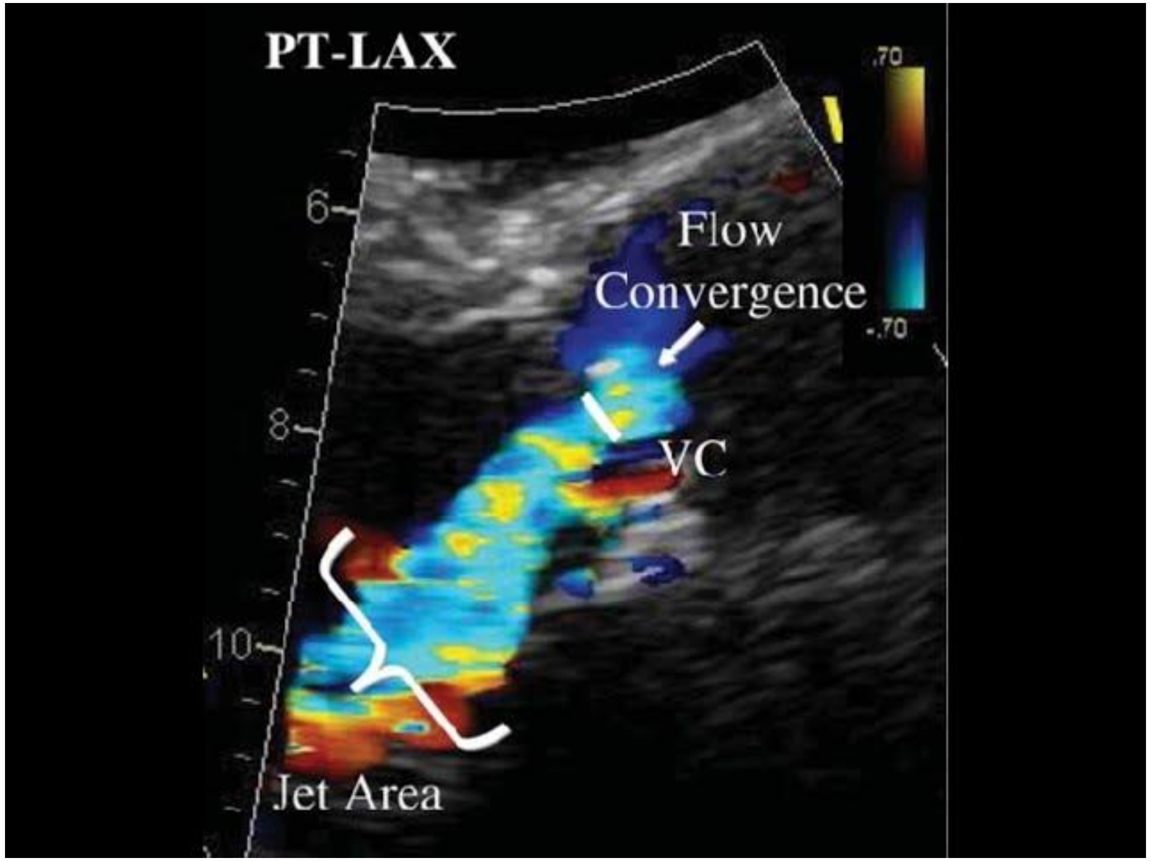


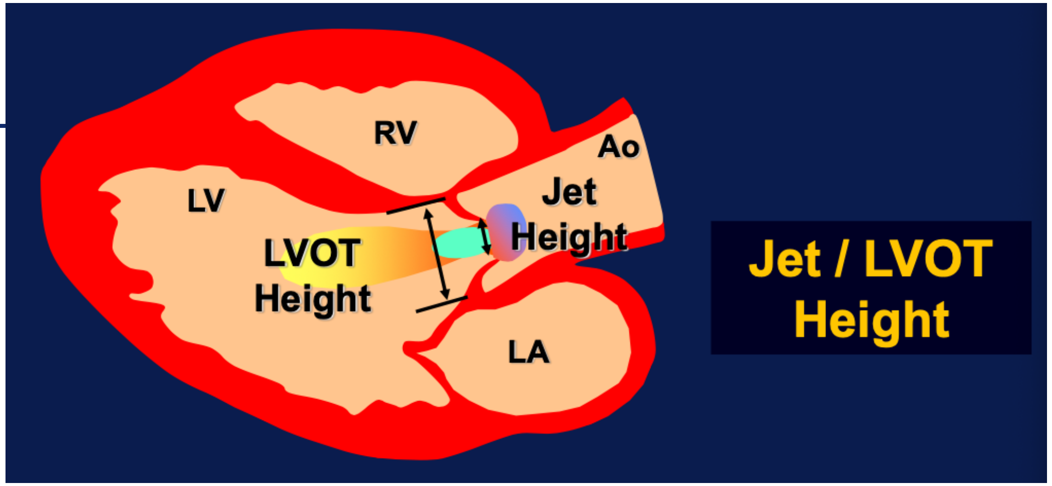
Vmax 0.51 m/s
Vmean 0.35 m/s
Pmax 1.04 mmHg
Pmean 0.51 mmHg
Env.Ti 689 ms
VTI 24.1 cm
HR 63 BPM
Vmax 1.65 m/s
Vmean 1.00 m/s
Pmax 10.93 mmHg
Pmean 5.12 mmHg
Env.Ti 294 ms
VTI 29.4 cm
HR 61 BPM
V
[m/s]
61
HR

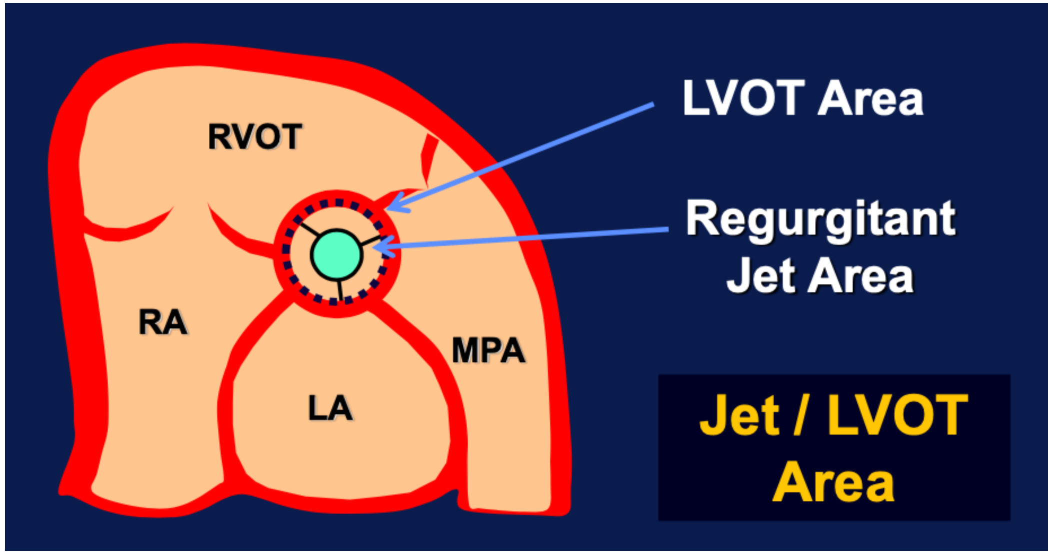


Freq.: 1.7 MHz/3.3 MHz
V
[cm]
66.67 mm/s
64
HR

Semiquantitative Parameters







	AR severity			
Semiquantitative parameters[§]	Mild	Moderate		Severe
VCW (cm)	<0.3	0.3-0.6		>0.6
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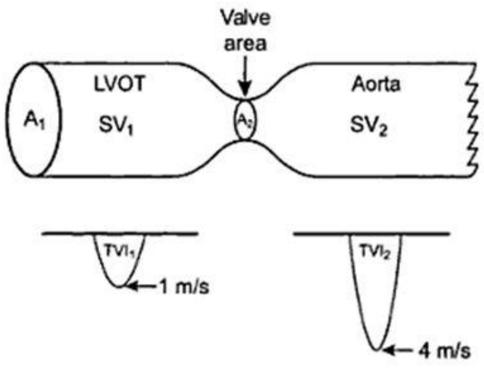
Jet CSA/LVOT CSA, central jets (%)	<5	5-20	21-59	≥60

Continuity Method

- Blood is not compressible
- Volume flowing through one valve is equal to the volume through all others

Continuity equation



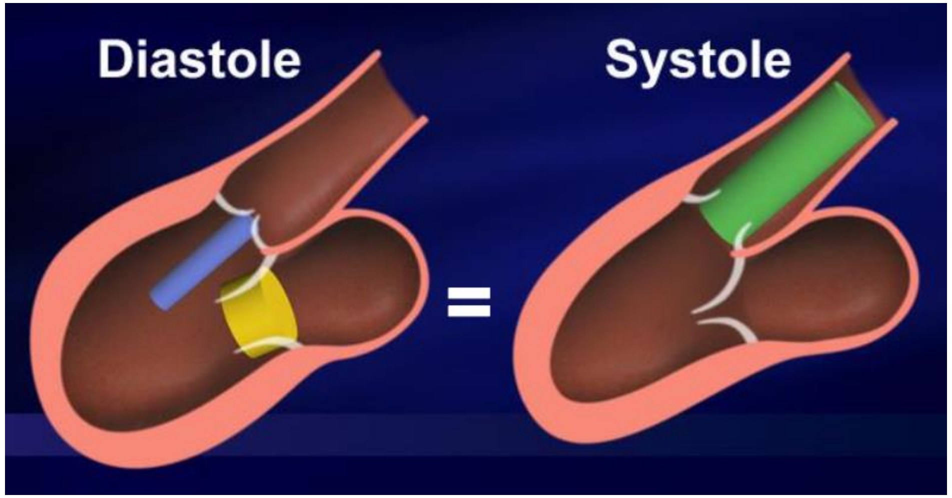
- $Q_{LVOT} = Q_{\text{thru all chambers}}$
- $A_{LVOT} \times VTI_{LVOT} = A_{point} \times VTI_{point}$
- $A_{point} = A_{LVOT} \times \frac{VTI_{LVOT}}{VTI_{point}}$

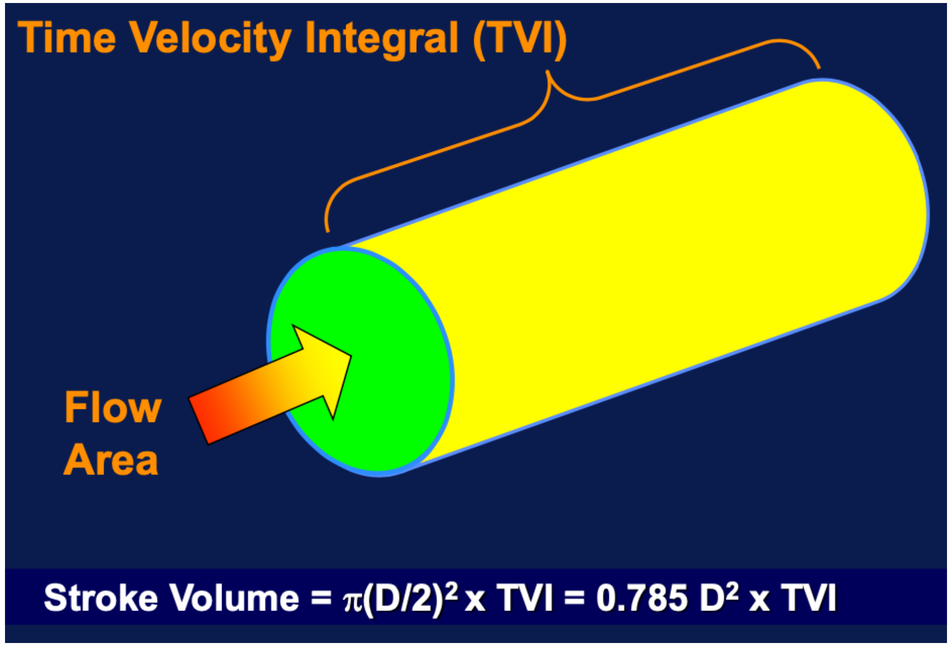
$$= \frac{\pi D^2}{4} \times \frac{VTI_{LVOT}}{VTI_{point}}$$

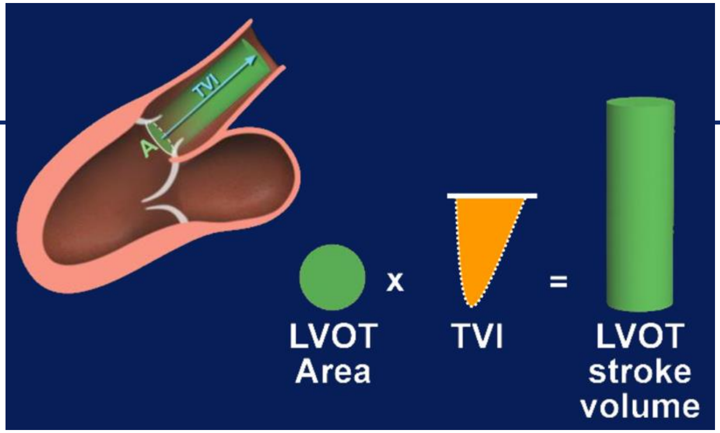
$$= 0.785\, D^2 \times \frac{VTI_{LVOT}}{VTI_{point}}$$

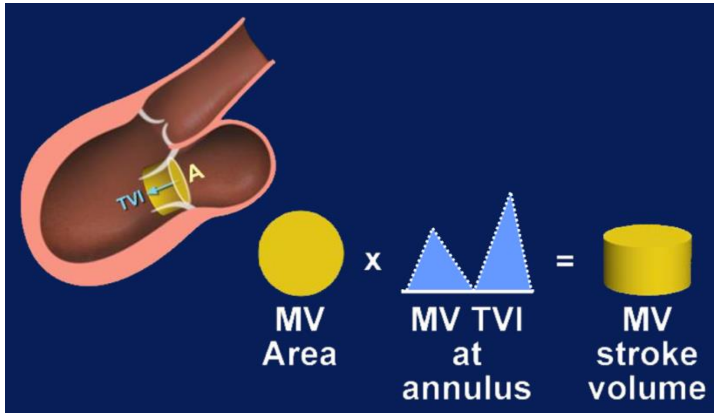
- No intracardiac shunts between the two points
- No significant regurgitant flow

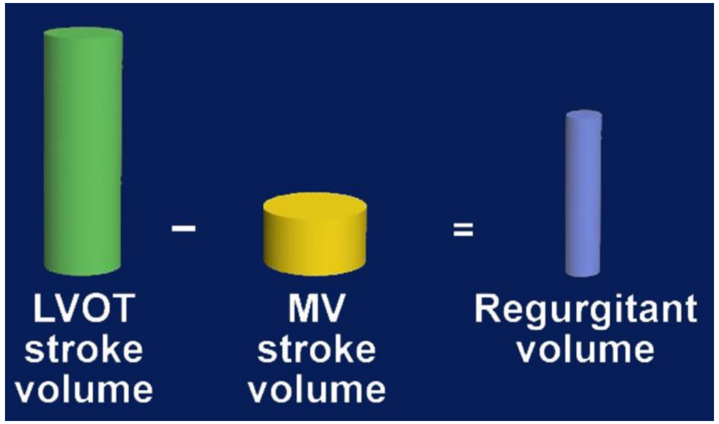
Continuity Method

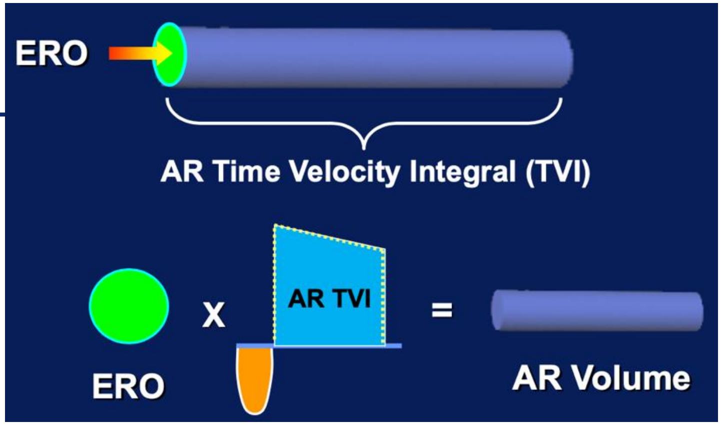


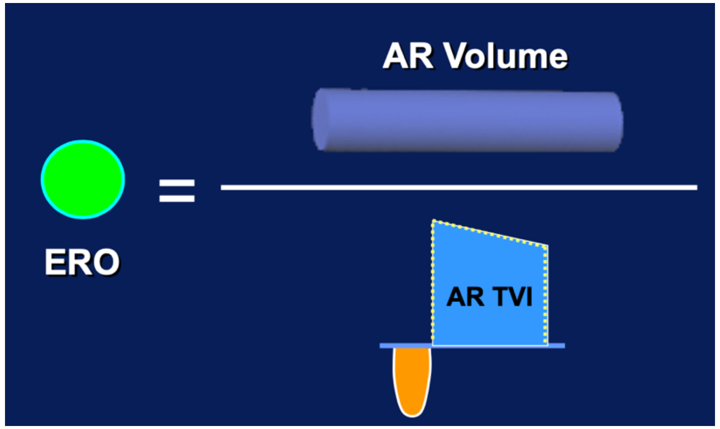


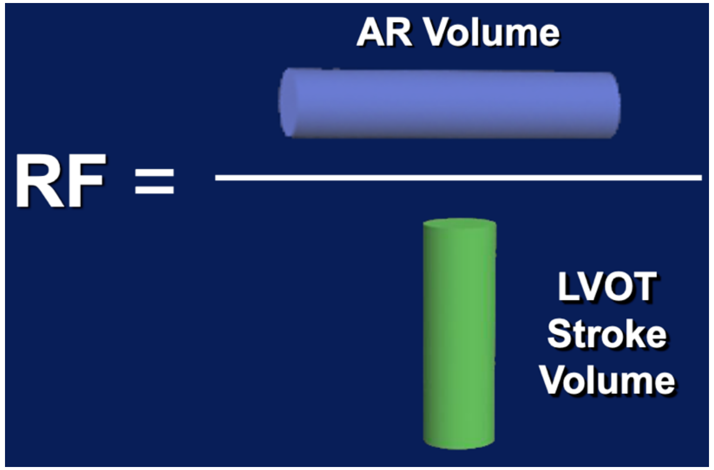


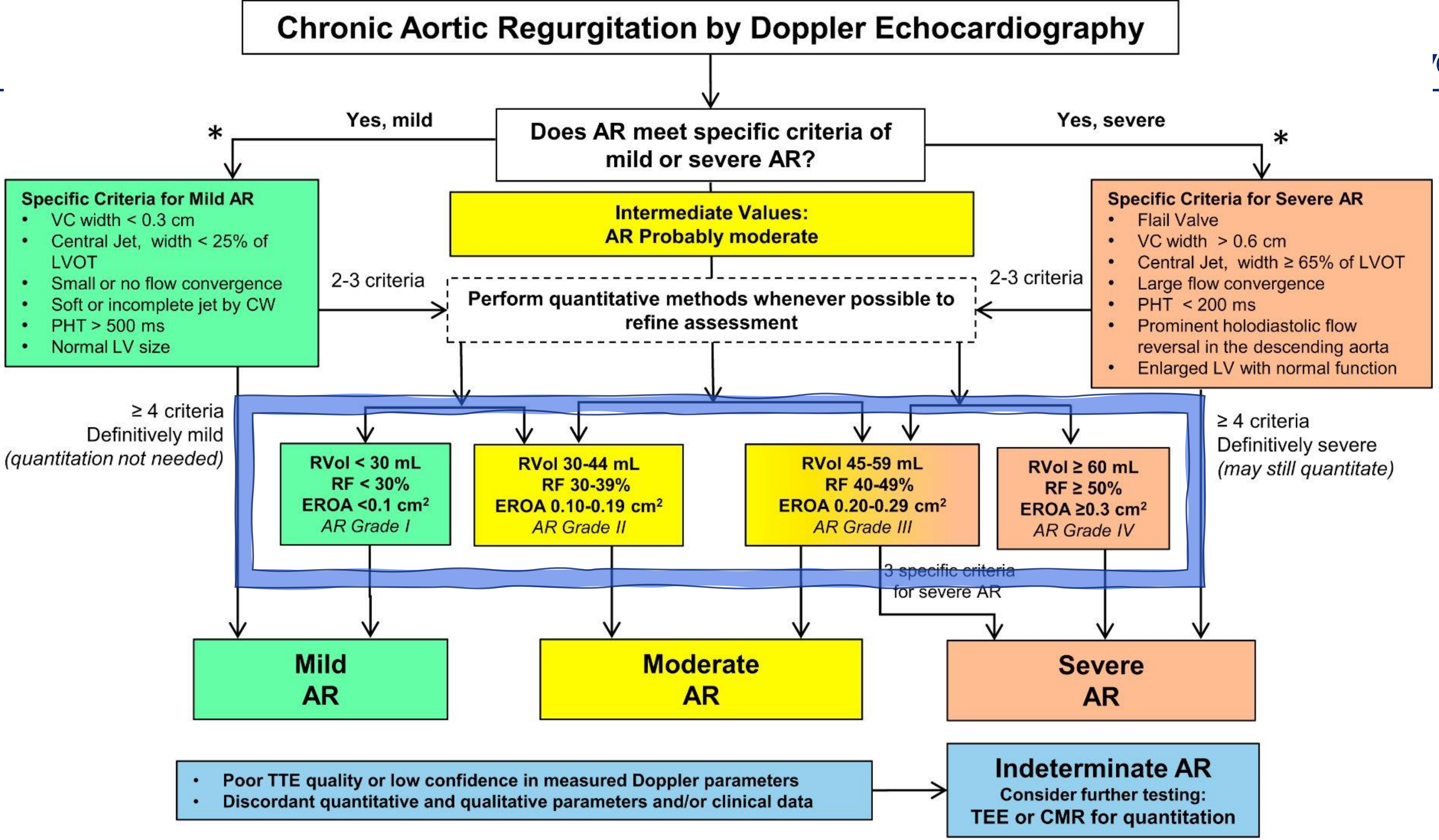












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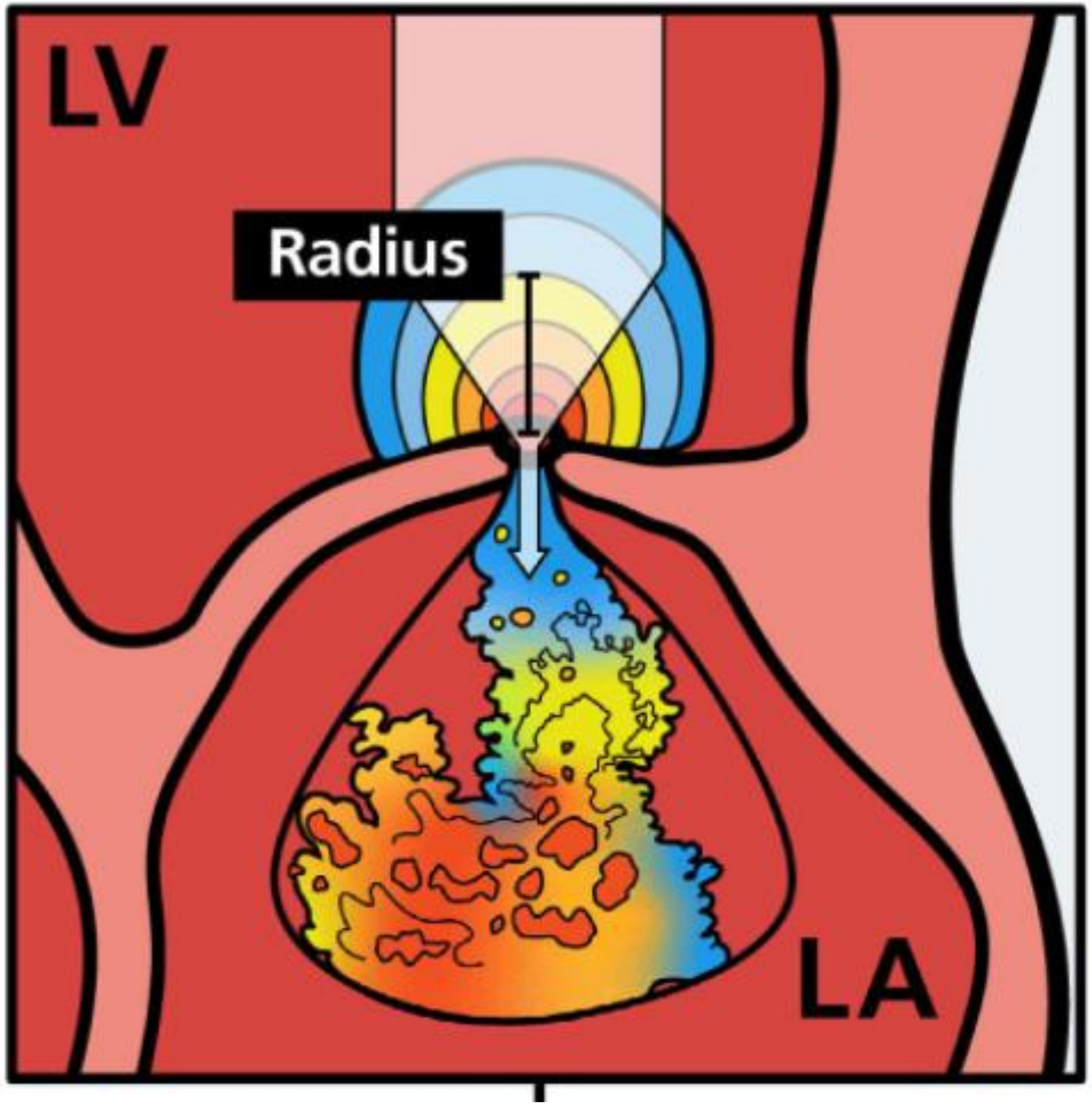
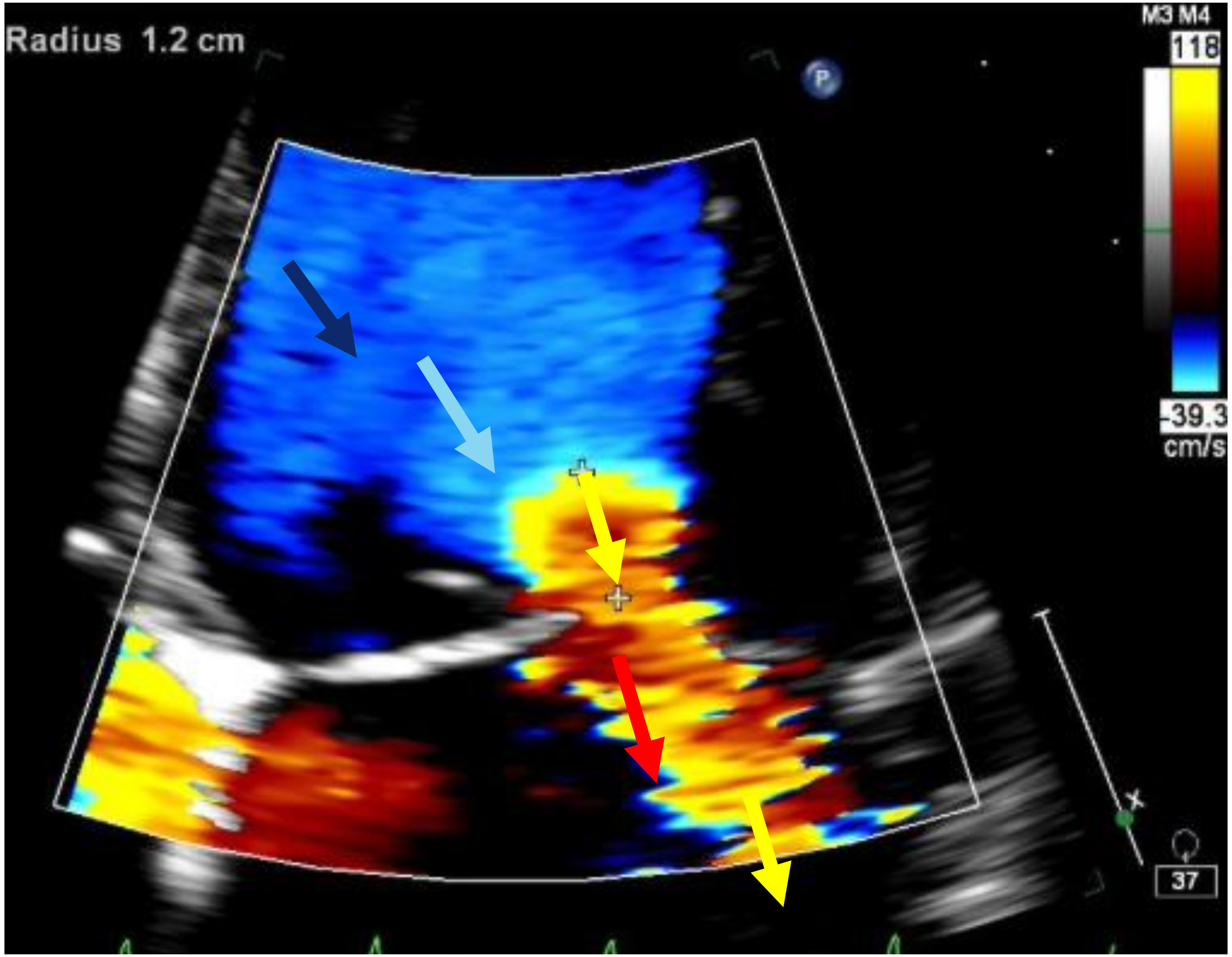
Continuity Method Potential Pitfalls

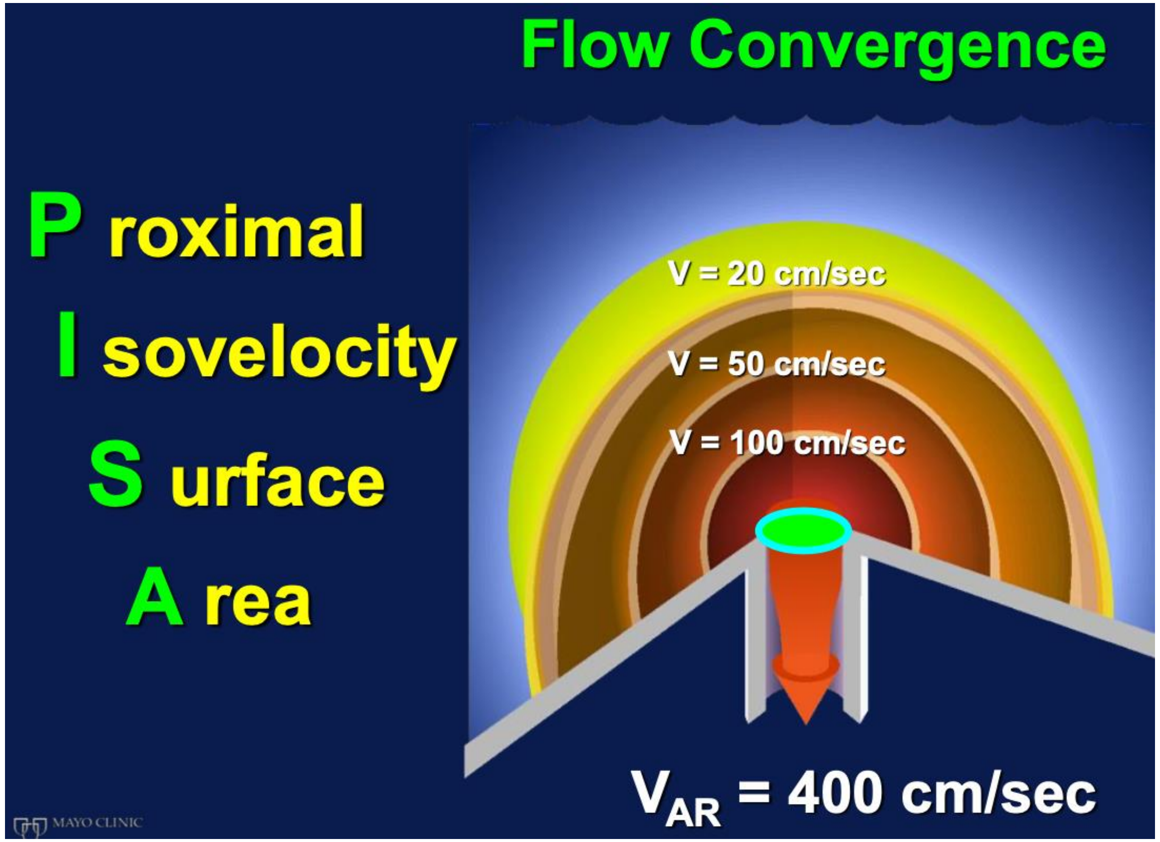
- Incorrect Doppler alignment to flow (theta > 20º)
- Incorrect sample volume placement (Place at annulus, not leaflet tips)
- Incorrect annular measurement: $(\text{error})^2$, Mitral annular calcification (MAC)
- Failure to trace modal velocity (especially MV)
- Geometric assumptions of circular annulus (LVOT - good, MV - fair, TV - poor)
- Mitral regurgitation > mild
- Arrhythmia; inadequate data averaged (use at least 5-8 cycles for Afib)

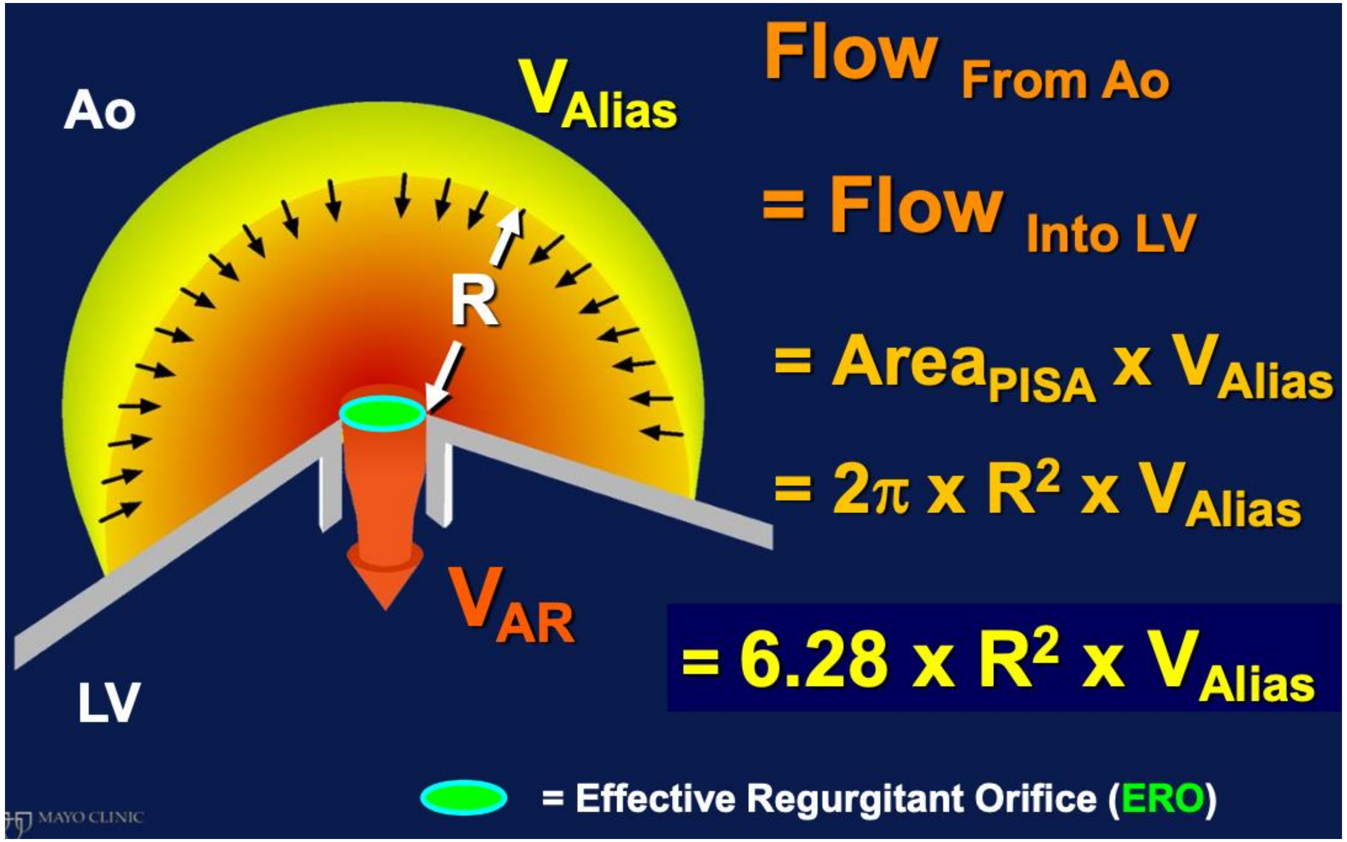
Flow Convergence

- Conservation of mass/volume




Radius 1.2 cm
M3 M4
118
-39.3
cm/s
37

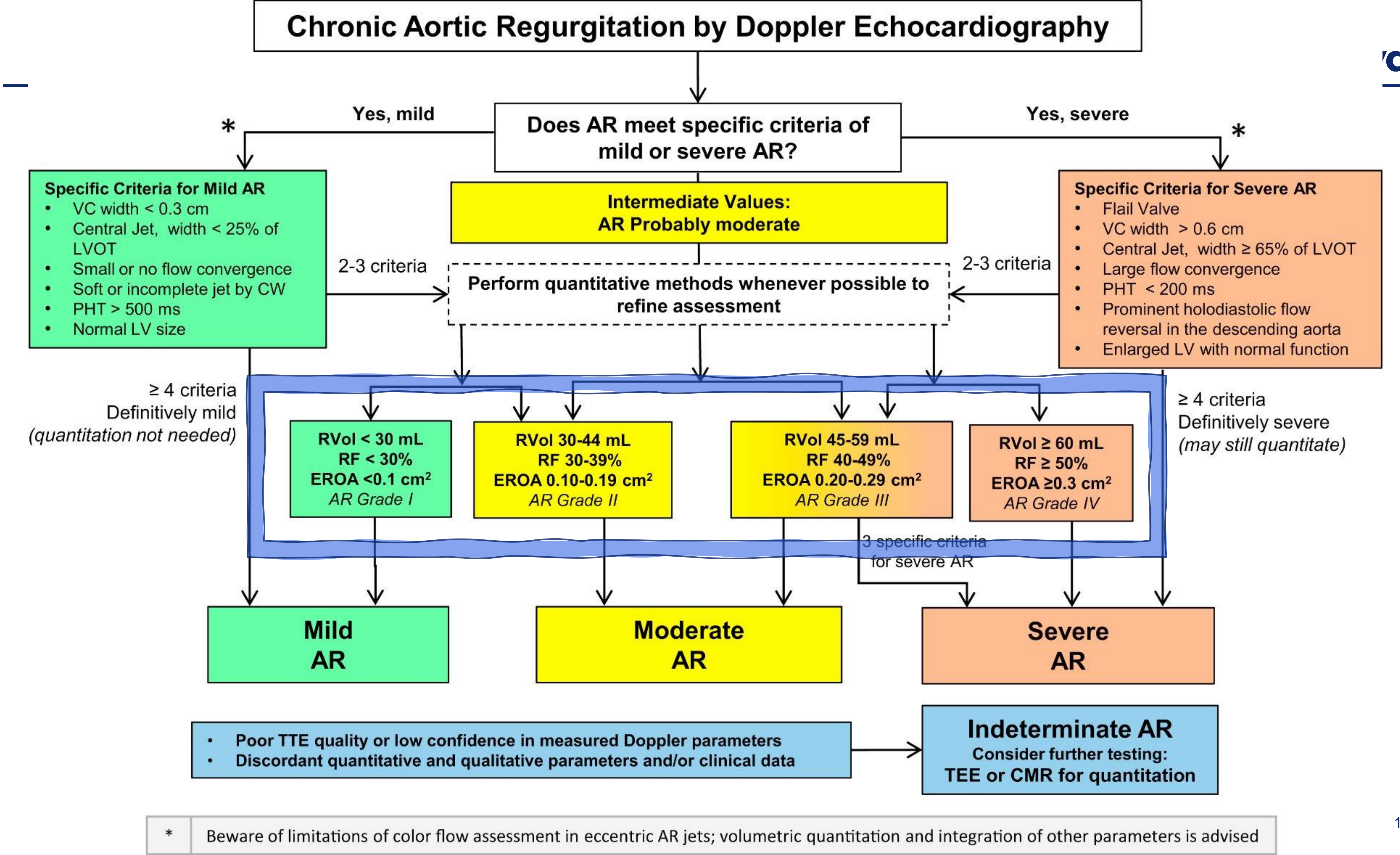




$$ERO = \frac{Flow_{AR}}{Velocity_{AR}}$$

Vmax

$$Volume_{AR} = ERO \times TVI_{AR}$$


Chronic Aortic Regurgitation by Doppler Echocardiography
Does AR meet specific criteria of mild or severe AR?
Yes, mild
*
Yes, severe
*
Specific Criteria for Mild AR
• VC width < 0.3 cm
• Central Jet, width < 25% of LVOT
• Small or no flow convergence
• Soft or incomplete jet by CW
• PHT > 500 ms
• Normal LV size
Intermediate Values: AR Probably moderate
Specific Criteria for Severe AR
• Flail Valve
• VC width > 0.6 cm
• Central Jet, width ≥ 65% of LVOT
• Large flow convergence
• PHT < 200 ms
• Prominent holodiastolic flow reversal in the descending aorta
• Enlarged LV with normal function
2-3 criteria
Perform quantitative methods whenever possible to refine assessment
2-3 criteria
≥ 4 criteria Definitively mild (quantitation not needed)
RVol < 30 mL RF < 30% EROA <0.1 cm²
AR Grade I
RVol 30-44 mL RF 30-39% EROA 0.10-0.19 cm²
AR Grade II
RVol 45-59 mL RF 40-49% EROA 0.20-0.29 cm²
AR Grade III
RVol ≥ 60 mL RF ≥ 50% EROA ≥0.3 cm²
AR Grade IV
≥ 4 criteria Definitively severe (may still quantitate)
3 specific criteria for severe AR
Mild AR
Moderate AR
Severe AR
• Poor TTE quality or low confidence in measured Doppler parameters
• Discordant quantitative and qualitative parameters and/or clinical data
Indeterminate AR
Consider further testing:
TEE or CMR for quantitation
* Beware of limitations of color flow assessment in eccentric AR jets; volumetric quantitation and integration of other parameters is advised

Advantages of PISA Method

Depends essentially on single measurement (aliasing radius) since v_{max} is relatively constant (and can even be estimated from SBP)
Less subject to propagation of errors than volumetric approach Uses axial resolution which is finer than lateral resolution used by vena contracta method
Well-suited for rapid quantification during interventions Less dependent on BP and instrument factors than jet area

Disadvantages of PISA Method

[Need to finish this]
Depends essentially on single measurement (aliasing radius) since v_{max} is relatively constant (and can even be estimated from SBP)
Less subject to propagation of errors than volumetric approach Uses axial resolution which is finer than lateral resolution used by vena contracta method
Well-suited for rapid quantification during interventions
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Thank you

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